

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460946.7901 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1747 +/- 0.0089
Transit depth (Rp/Rs)^2	3.05 +/- 0.31 [%]
Orbital Inclination (inc)	87.74 +/- 1.06
Airmass coefficient 1 (a1)	1.064 +/- 0.015
Airmass coefficient 2 (a2)	-0.052 +/- 0.025
Scatter in the residuals of the lightcurve fit is	1.42 %
Best Comparison Star	#2 - [371, 333]
Optimal Aperture	4.02
Optimal Annulus	12.72
Transit Duration (day)	0.0857 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1747 $\pm$ 0.0089 vs 0.1646 (deviation 0.95 σ)
Duration fit vs published	0.0857 d vs 0.0758 d (deviation 2.96 σ)
Mid-time O-C	4.5 min
Depth SNR	16.5
Residual scatter	1.42 %

Light curve

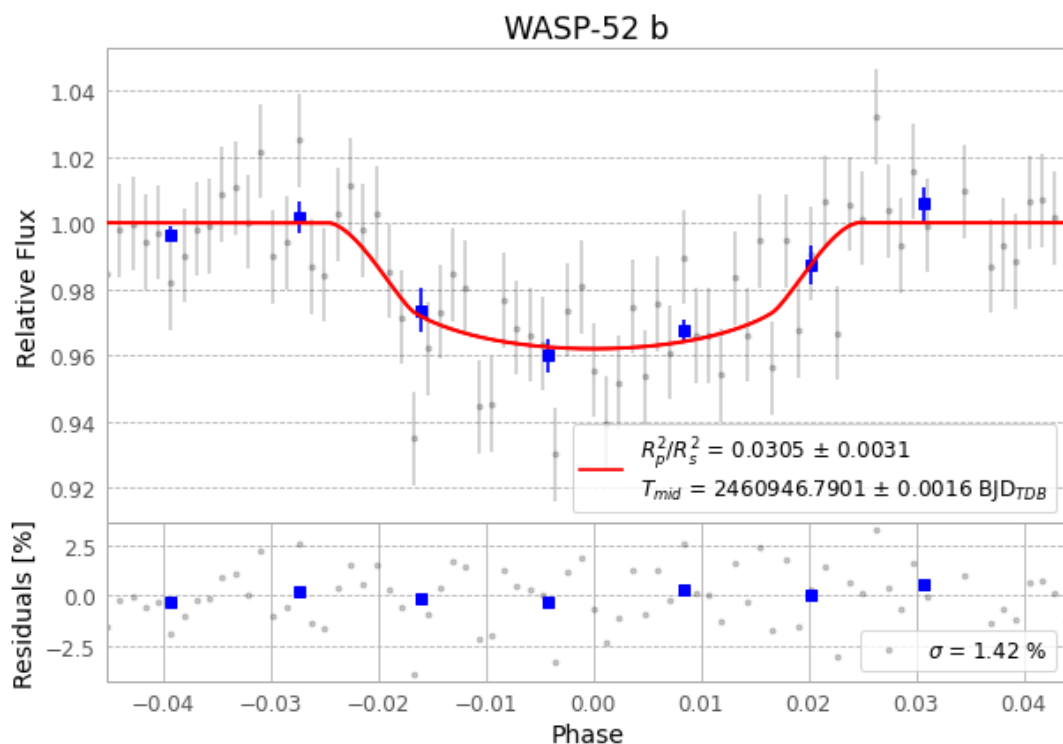


Figure 1 — FinalLightCurve_WASP-52 b_2025-09-27.png (SHA-256 0c17b2334855f8c3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cd5eb1fb15d9d376...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

76 FITS frames (50 MB), WASP-52250928045415.FITS → WASP-52250928083915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-250928.log (c760ba56e469...), FinalLightCurve_WASP-52 b_2025-09-27.png (0c17b2334855...), FinalLightCurve_WASP-52 b_2025-09-27.csv (e396c88c1442...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 19:06Z.